

Electromagnetic Forces On Coils And Bars Inside The Slot of Hydro-Generator

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Abstract— The coils and bars installed in the stator core are subjected to transversal electromagnetic (EM) field inside the stator core slot that in turn, develops electromagnetic forces between the top and bottom layers. This subject matter and the theoretical background have been discussed in various academic papers in the past and is well known among equipment manufacturers. It is not, however, well known among plant engineers. There is a misconception about the amplitude and direction of those forces and how they are applied to hydro-generators stator winding.

An electromagnetic finite element analysis (FEA) was performed during a recent project to study the effect of electromagnetic forces on stator and rotor. A detailed model of the stator and rotor was prepared and the electromagnetic forces on the top and bottom bars were calculated and analyzed. This analysis demonstrated how saturation of the stator teeth, actual magnetic field distribution inside the slot, eddy current in the damper bars and the shape of the salient poles impacted the electromagnetic forces in the slot.

This paper summarizes the theory and basics of EM forces on conductors inside the core slot, provides calculation of the electromagnetic forces at rated operating conditions and fault conditions as function of the phases on the top and bottom bar, and provides a comparison of the forces calculated using an electromagnetic FEA. Furthermore, the forces inside the slot are discussed, which is an important factor for securing the winding inside the slot.

Keywords—Lorentz forces, radial forces, tangential forces, slot wedges, side packing, top ripple springs

I. INTRODUCTION

The analysis of the electromagnetic forces in the slot was developed for a three-phase hydro-generator. Nevertheless, the results are applicable to any three-phase alternating current machine.

The forces in the top and bottom layers are caused by the current flowing on the top and bottom layers of the coils or bars. The force depends upon the amplitude of the current and the phase relationship between the top and bottom layer. Contrary to common belief, the forces in the slot are mostly towards the bottom of the slot, and a relatively small amplitude is applied to the slot wedges. A conservative analytical calculation was performed to determine the amplitude of the electromagnetic forces developed in the slot area.

There are several factors that impact the calculation of the forces inside the slot, for instance saturation of the slot teeth, the actual magnetic field distribution inside the slot, eddy current in the damper bar and the shape of the salient pole. The influence of those factors on the force amplitude is analyzed using a time stepped electromagnetic FEA performed at full load and compared with the theoretical values.

II. ANALYTICAL CALCULATION OF FORCES INSIDE THE SLOT

The current flowing on conductor inside a slot creates magnetic field in the iron portion around the slot and the magnetic circuit is closed by the magnetic field going from one slot wall to the other slot wall as shown on Fig. 1. As the magnetic field in the slot is in tangential direction it is commonly known as “transversal field”. Assuming an infinite permeable stator core iron and straight transversal field lines, the forces in the conductors will only have radial component. The distributed current interacts with its own transversal magnetic field in the slot and results in forces in radial direction. As the strands are bundled together in top and bottom conductors, a uniform current density is assumed as shown in Fig. 1.

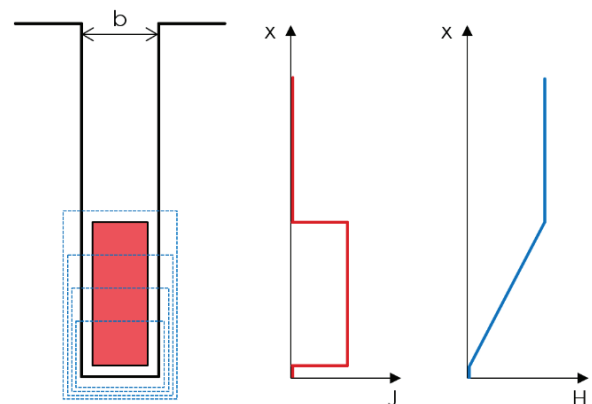


Fig. 1. Current density and magnetic field strength in the slot for a current carrying conductor

The resulting magnetic flux density depends on the permeability of the magnetic material. With increased iron saturation, the relative permeability of the iron decreases, which results in reduction of magnetic flux density in the slot. This means that if the saturation is not considered, the calculated magnetic flux density and expected forces on the bars will be larger, which is a more conservative approach.

The resulting force is calculated by integrating vector cross product of the current density, which can be considered evenly distributed in the conductor cross section, and the magnetic flux density over the volume of the bar [1],

$$\vec{F} = \iiint \vec{J} \times \vec{B} dV \quad (1)$$

Below is the resulting force of the volume integral for the top and bottom conductors separately from [2].

$$\vec{F}_{bottom} = \frac{\mu_0 l_z}{2b} i_{bottom}^2 \vec{e}_r \quad (2)$$

$$\vec{F}_{top} = \frac{\mu_0 l_z}{2b} i_{top} \left(i_{bottom} + \frac{1}{2} i_{top} \right) \vec{e}_r \quad (3)$$

In the above expressions, μ_0 is the permeability of free space, l_z is the axial length of the conductor inside the slot, b is the slot width and $\vec{e}_r$ is the normal radial vector pointing toward slot bottom. The force on the bottom conductor only depends on the current of the bottom conductor and its direction is always toward the bottom of the slot. The amplitude and the direction of the force on the top conductor depends on the current of both top and bottom conductors.

If both conductors belong to the same phase, then the direction of the force is always towards the bottom of the slot. If the conductors belong to two different phases, there are times that the top conductor force has a negative value, meaning the top conductor applies force against the slot wedge. Although, most of the time, the direction of the top conductor force is towards the bottom of the slot and the largest forces are also in this direction.

For a given current on the bottom conductor (i_{bottom}) and slot width (b) the magnitude of the force per unit length in the slot can be calculated using following expression:

$$\frac{F_{bottom}}{l_z} = \frac{6.2832 \cdot 10^{-7}}{b} i_{bottom}^2 \quad (4)$$

The unit of above force per unit of length is N/m.

The frequency is twice rated frequency.

The magnitude of all other forces inside the slot, can be calculated as a function of the bottom conductor force using pre-determined multiplication factors as follows:

$$F = k_f F_{bottom} \quad (5)$$

For instance, the amplitude of the forces on the slot content for a winding with 2500 A rms per bar, slot width of 0.9 inch would be using (4):

$$F_{bottom} = 344 \text{ N/m}$$

III. CASE 1

The case 1 consists of a hydro-generator with following characteristics:

Rated Output:	95.5 MVA
Rated Voltage:	13,800 V
Rated Current:	3,995 A
Number of slots:	432
Number of Circuits:	3
Slots per pole and phase:	$1 \frac{5}{7}$
Winding type:	Coils with two turns
Slot Width:	24.76 mm (0.975 inch)
Core Length:	1460.5 mm (57.5 inches)

The amplitude of the forces per unit of length on the bottom coil would be using (4):

$$\frac{F_{bottom}}{l_z} = 360 \text{ N/m}$$

TABLE 1: k_f FACTOR

LOCATION	DIRECTION	SAME PHASE	DIFFERENT PHASE
Bottom Conductor	Towards the bottom of slot	1	1
Top Conductor	Towards the bottom of slot	3	2.323
Top Conductor	Towards the slot wedge	0	0.323
Bottom Slot	Towards the bottom of slot	4	3
Center filler	Towards the bottom of slot	3	2.323
Slot Wedge	Towards the air gap	0	0.323

Using k_f factors provided on Table 1, and the core length provide above, the forces on the slot content can be calculated. The amplitudes of the force are provided on Table 2.

TABLE 2: Amplitude of the forces inside the slot for Case 1

LOCATION	DIRECTION	Same Phase (N)	Different Phases (N)
Bottom Coil	Towards the bottom of slot	526	526
Top Coil	Towards the bottom of slot	1578	1222
Top Coil	Towards the slot wedge	0	170
Bottom of Slot	Towards the bottom of slot	2104	1578
Center Filler	Towards the bottom of slot	1578	1221
Slot Wedge	Towards the air gap	0	170

The relatively small amplitude of the force on the slot wedge towards the air gap is highlighted below.

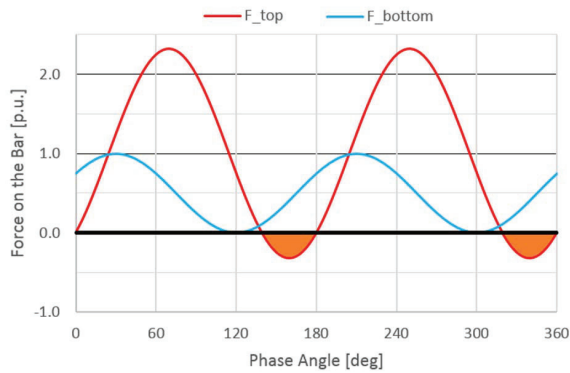


Fig. 2. Force on the conductors from different phases

Fig 2 indicates that the force is positive most of the time, meaning the direction of the force is towards the bottom of the slot. But on top coil a force equal to 32.3% of the maximum bottom bar force occurs on the direction of the slot wedge.

A. Fault Event

The maximum force is produced during the fault events when the amplitude of the fault current become very high and non-symmetric, e.g. three phase short circuit. During a short circuit the currents can be as large as the 10 times of the rated current. The forces are proportional to the square of current, therefore, the force in the slot may become as high as 100 times of the forces during the normal operation.

In particular, the force on the slot wedge can increase to 32.3 times the force on the bottom coil. Using the Table 2 values, the top coil force on the slot wedge would be 17000 N and the top bar force toward the bottom of the slot would be 122100 N. This alternating force could potentially cause the top coil to vibrate inside the slot and could overstress the slot wedge.

In the past, the slot wedges were installed with hard filler and a relatively common failure was loose slot wedges. The wedges

were tightly driven over top fillers, however due to mechanical vibration and constant alternating radial forces inside the slot, the fillers may relax overtime or move and settle causing the wedges to become loose. The loose wedge may allow the top coil to vibrate inside the slot causing damage to the slot corona protection and slot ground wall insulation. Nowadays the use of top ripple springs reduces drastically the chance of top coils to become loose inside the slot. The top ripple spring compression force is typically several folds higher than the electromagnetic forces and will always keep positive pressure on the coil. Although the bar forces alternate directions, the force on the wedge with top ripple spring will be always in the same direction. There is an equipment manufacturer that does not use top ripple springs, instead injects a conformable compound underneath the slot wedge through radial holes on the wedges. The equipment manufacturer reports successful results with no loose slot wedges.

IV. CASE 1- 2D TIME STEPPED FEA

In reality, there are multiple factors that have influence on the magnetic field in the slot. For instance, the stator core tooth is subjected to saturation, the shape of the rotor pole tip will cause non-uniform flux distribution, and damper bar currents are not uniform on all damper bars. The question becomes: how much do those factors affect the bar forces in the slot in comparison with the simplified and conservative analytical calculation?

A 2D time stepped electromagnetic finite element analysis was performed to investigate the contribution of those factors on the forces.

The software Flux 2D from Altair was used for this investigation.

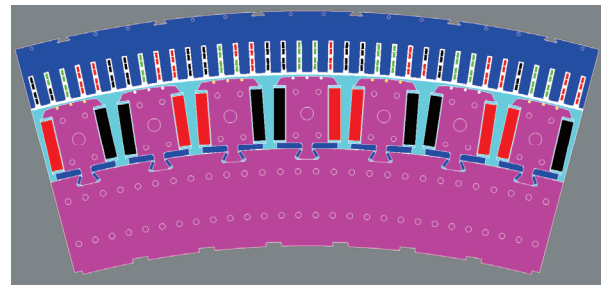


Fig. 3. Case 1- Study Domain

The study domain was modeled and consisted of a section with 7 poles (Fig. 3). The sides of the model are subjected to periodicity boundary condition with anticyclic conditions.

A relatively fine mesh was used, in particular on the damper bars and slot wedges. The total number of nodes exceeded 295,000 using with 2nd order finite elements. Fig. 4 shows a detail of the mesh around a part of the airgap.

External circuit connections were used on the stator winding, field winding and damper winding. Each winding was represented with its inductance and resistance.

Several iterations were run to calibrate the magnetic properties of the stator and rotor material such that the voltage

induced at the generator terminal at several field current and saturation level match the actual open circuit saturation curve of the generator.

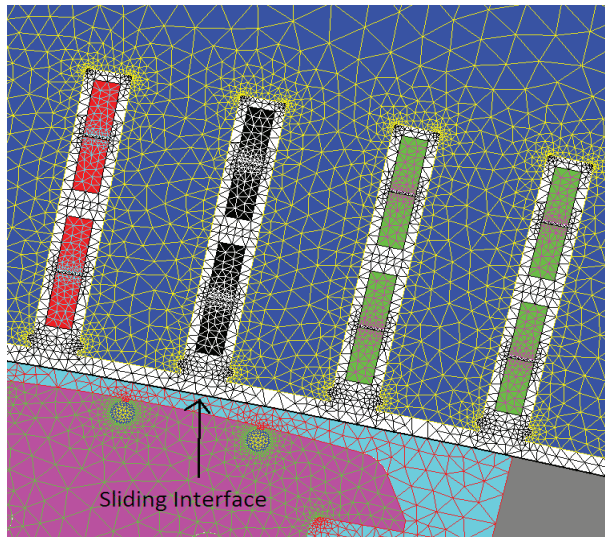


Fig. 4. Case 1- Mesh Details

With a calibrated model, the study was performed at no-load and full load conditions.

One outcome of the FEA analysis at full load conditions was the electromagnetic forces on the top and bottom coil. Two slots were selected, one with the top and bottom coil belonging to the same phase and another with different phases on the top and bottom coil.

The Lorenz Force was calculated by integrating the product of the current density vector and magnetic flux density over the copper cross section and length.

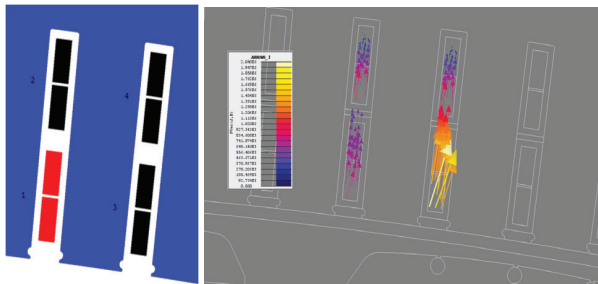
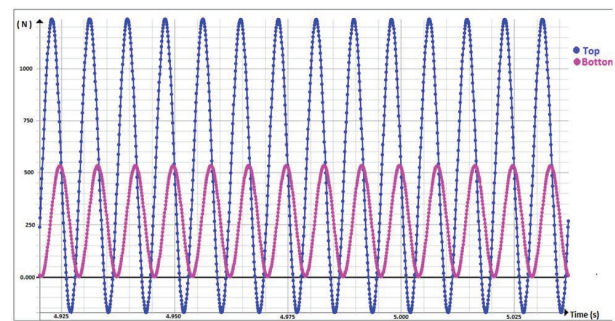


Fig. 5. Slot filling and forces on the top and bottom coil

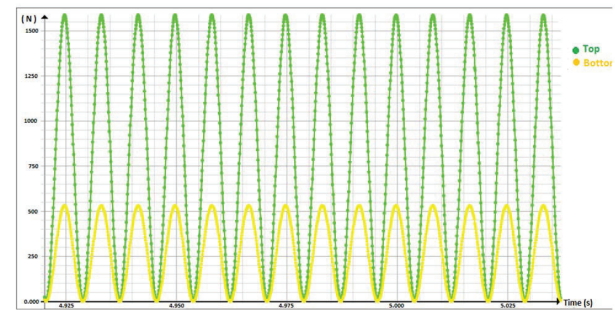
Fig. 5 shows the forces vector on the top and bottom coil at 5.0329 seconds. The slot to the left had two different phases and the slot to the right had the same phase on the top and bottom coils.

As predicted by the theory, the direction of the forces faces toward the slot bottom.

The finite element analysis shows that there are radial and tangential components on the bar forces. Each component was calculated separately to be compared to the analytical calculation.



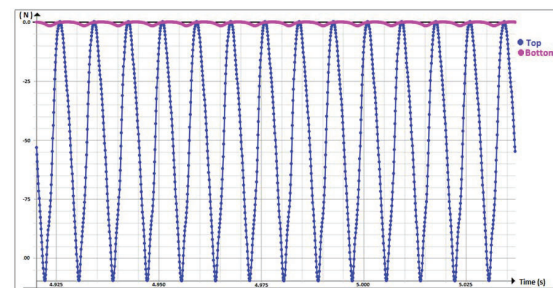
Radial forces for slot with same phases on top and bottom coils



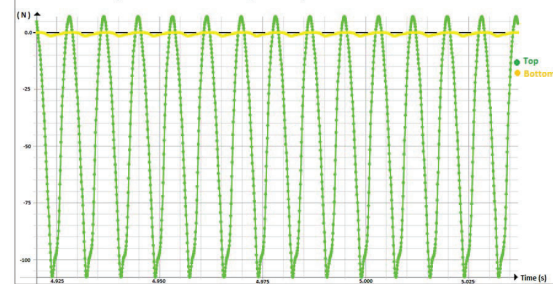
Radial forces for slot with different phases on top and bottom coil

Fig. 6. Radial Forces on top and bottom coil

Fig. 6 shows the radial component of the force on the top and bottom coil. The upper chart shows the force for slots with two different phases. And the lower chart shows the forces for slot with the same phase.



Tangential forces for slot with same phases on top and bottom coils



Tangential forces for slot with different phases on top and bottom coil

Fig. 7. Tangential Forces on top and bottom coil

Fig. 7 shows the tangential component of the force on the top and bottom coil. The tangential force on the bottom coil is very

small. The tangential force on the top coil is substantial lower than the radial forces.

The tangential force for slots with the same phase on the top and bottom coil reverses direction.

In the past, stator coils were installed with hard side filler on one side of the coils. A typically failure of this packing process was migration of side filler out the slot, mechanical erosion of the coil insulation and displacement of the coil. Nowadays, stator coils are installed either with side ripple springs or conformable filler material. The side ripple spring will maintain positive pressure on the coil side preventing the coil from vibrating. Conformable filler materials will fill the gap between the coil and the slot wall and lock the coil inside the slot.

TABLE 3: Comparison of the radial force amplitude calculated analytically and by FEA for Case 1

LOCATION	DIRECTION	Same Phase (N)		Different Phases (N)	
		Analytical	FEA	Analytical	FEA
Bottom Bar	Radially towards the bottom of slot	526	530	526	534
Top Bar	Radially towards the bottom of slot	1578	1587	1222	1237
Top Bar	Radially towards the slot wedge	0	0	170	172

TABLE 4: Comparison of the tangential force amplitude calculated analytically and by FEA for Case 1

LOCATION	DIRECTION	Same Phase (N)		Different Phases (N)	
		Analytical	FEA	Analytical	FEA
Bottom Bar	Tangentially clockwise	0	2	0	2
Top Bar	Tangentially clockwise	0	108	0	109
Top Bar	Tangentially Counter clockwise	0	7	0	0

V. CASE 2

The case 2 consists of a hydro-generator with following characteristics:

Rated Output:	95.5 MVA
Rated Voltage:	13,800 V
Rated Current:	3,995 A
Number of slots:	576
Number of Circuits:	2
Slots per pole and phase:	$2 \frac{2}{7}$
Winding type:	Roebel Bar
Slot Width:	23.19 mm (0.913 inch)
Core Length:	1447.8 mm (57 inches)

The amplitude of the forces on the bottom bar would be using (4):

$$\frac{F_{bottom}}{l_z} = 216 \text{ N/m}$$

Using k_f factor provided on Table 1 and the core length provided above, the amplitude of the forces on the slot content can be calculated. The amplitudes of the force are provided on Table 5.

TABLE 5: Amplitude of the forces inside the slot for Case 2

	LOCATION	DIRECTION	Same Phase (N)	Different Phases (N)
1	Bottom Bar	Towards the bottom of slot	313	313
2	Top Bar	Towards the bottom of slot	940	728
3	Top Bar	Towards the slot wedge	0	101
4	Bottom of Slot	Towards the bottom of slot	1253	940
5	Center Filler	Towards the bottom of slot	940	728
6	Slot Wedge	Towards the air gap	0	101

VI. CASE 2- 2D TIME STEPPED FEA

Similarly to case 1, a 2D time stepped electromagnetic finite element analysis was performed to investigate the contribution of the stator core teeth saturation, the shape of the rotor pole tip, and the non-uniform damper bar currents on the radial and tangential forces.

The software Flux 2D from Altair was used for this investigation.

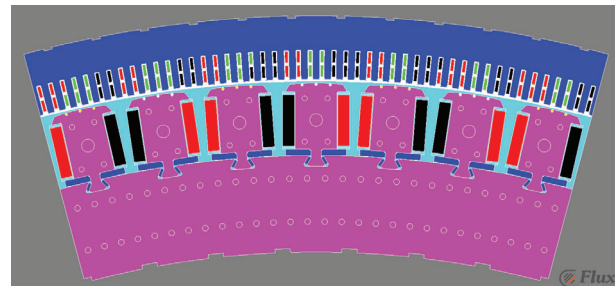


Fig. 8. Case 2- Study Domain

First the study domain was modeled and consisted of a section with 7 poles (Fig. 8). The side of the model are subjected to periodicity boundary condition with anticyclic conditions.

A relatively fine mesh was used in particular on the damper bars and slot wedges. The total number of nodes exceeded 365,000 using with 2nd order finite elements. The increased number of nodes in relation to case 1 was due to the increase of the number of slots in the study domain. Fig. 9 shows details of the meshing.

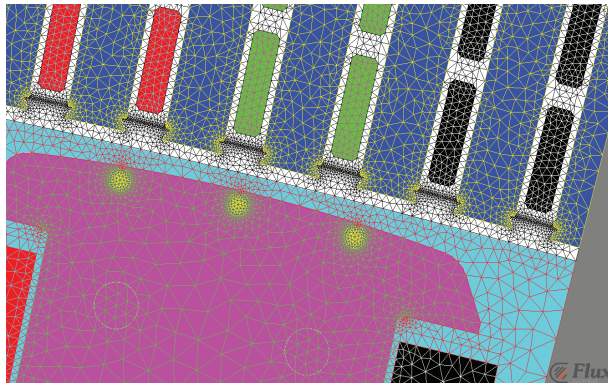


Fig. 9. Case 2- Mesh Details

Similarly to case 1, external circuit connections were used on the stator winding, field winding and damper winding. Each winding was represented with its inductance and resistance.

Several iterations were run to calibrate the magnetic properties of the stator and rotor material such that the voltage induced at the generator terminal at several field current and saturation level match the actual open circuit saturation curve of the generator.

With a calibrated model, the study was performed at no-load and full load conditions.

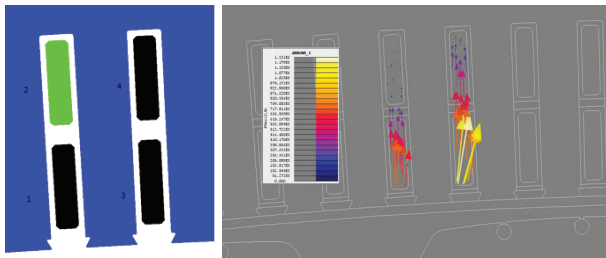


Fig. 10. Slot filling and forces on the top and bottom coil

Fig 10 shows the forces vector on the top and bottom bar at 5.03089 seconds. The slot to the left had two different phases and the slot to the right had the same phase on the top and bottom bar.

Similarly to case 1, the Lorenz Force was calculated by integrating the product of the current density vector and magnetic flux density over the copper cross section and length.

As predicted by the theory, the direction of the forces faces toward the slot bottom.

The finite element analysis shows that there are radial and tangential components on the bar forces. Each component was calculated separately to be compared to the analytical calculation.

Fig. 11 shows the radial component of the force on the top and bottom bar. The upper chart shows the force for slots with two different phases. And the lower chart shows the forces for slot with the same phase.

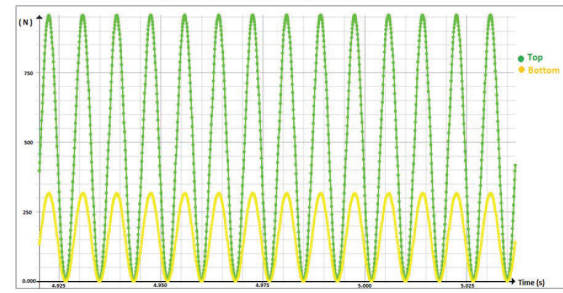
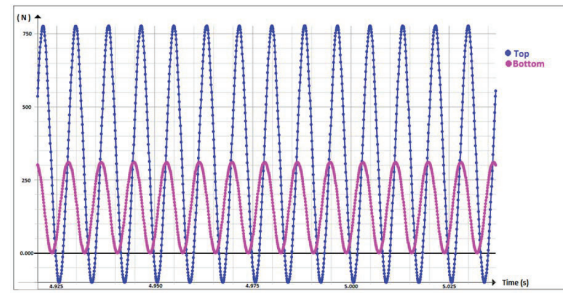


Fig. 11. Radial Forces on top and bottom coil

Figure 12 shows the tangential component of the force on the top and bottom bar.

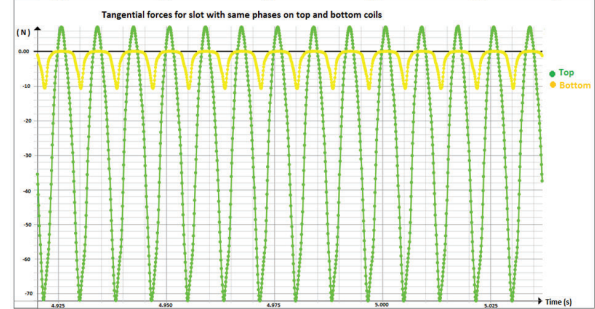
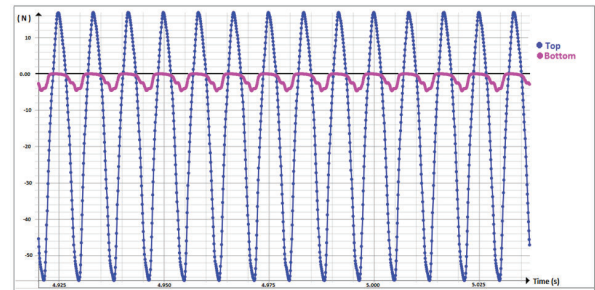


Fig 12: Tangential Forces on top and bottom coil

Tables 6 and 7 shows a comparison between the results of analytical calculation and FEA.

TABLE 6: Comparison of the radial force amplitude calculated analytically and by FEA for Case 2

LOCATION	DIRECTION	Same Phase (N)		Different Phases (N)	
		Analytical	FEA	Analytical	FEA
Bottom Bar	Radially towards the bottom of slot	313	297	313	310
Top Bar	Radially towards the bottom of slot	940	955	728	776
Top Bar	Radially towards the slot wedge	0	0	101	101

TABLE 7: Comparison of the tangential force amplitude calculated analytically and by FEA for Case 2

LOCATION	DIRECTION	Same Phase (N)		Different Phases (N)	
		Analytical	FEA	Analytical	FEA
Bottom Bar	Tangentially clockwise	0	11	0	5
Top Bar	Tangentially clockwise	0	72	0	57
Top Bar	Tangentially Counter-clockwise	0	7	0	17

VII. COMPARISON ANALITYCAL VS FEA

Table 3 shows a comparison between the amplitude of the radial force calculated analytically and by electromagnetic FEA for Case 1.

The results are very similar with maximum deviation of less than 1.5%. Therefore, there is very good concordance between the analytical calculation and the FEA for Case 1.

Table 4 shows the amplitude of the tangential force calculated by electromagnetic FEA for Case 1. The amplitude of the tangential force on the bottom coil is relatively small. The amplitude of the force on the top coil is several folds larger than the bottom coil force, although represents less the 9% of the radial force.

Table 6 shows a comparison between the amplitude of the radial force calculated analytically and by electromagnetic FEA for Case 2.

The results are very similar with maximum deviation of less than 6.5%. Therefore, there is good concordance between the analytical calculation and the FEA for Case 2.

Table 7 shows the amplitude of the tangential force calculated by electromagnetic FEA for Case 2. The amplitude of the tangential force on the bottom coil is relatively small. The amplitude of the force on the top coil although several folds larger than the bottom coil force, is less the 8% of the radial force.

VIII. CONCLUSIONS

The electromagnetic FEA confirmed that the radial forces on the conductors inside the slot are mostly towards the bottom of the slot and a relatively small portion of the force on the top conductor (32.3% of the force on the bottom conductor) is towards the slot wedges.

The FEA also shows that there is a tangential component acting mainly on the top conductor. The amplitude of the tangential is much smaller than the radial forces. The tangential force also reverse directions, indicating a potential risk of top conductor vibrating tangentially should the side packing become loose.

On both cases analyzed, there were good concordance between the analytical and the FEA results for the radial force.

During fault conditions, for instance a three phase short-circuit or out-of-phase synchronization, the fault currents could reach up to 10 times the rated value. These currents can result in forces that could be as high as 100 times of forces at rated conditions with alternating direction on top conductor, changing the direction between bottom of the slot and the slot wedge. This condition could lead to vibration of the top conductor inside the slot if it is not supported sufficiently inside the slot.

The use of top ripple springs as well as side ripple springs seem very effective to maintain positive pressure on the top conductor which keeps the direction of the force unchanged, and hence prevents the conductor from vibrating during fault conditions and normal operation.

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